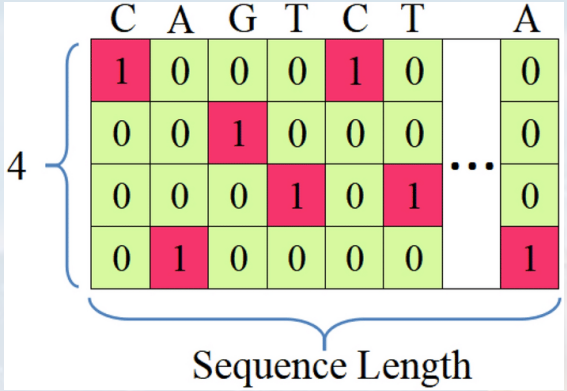
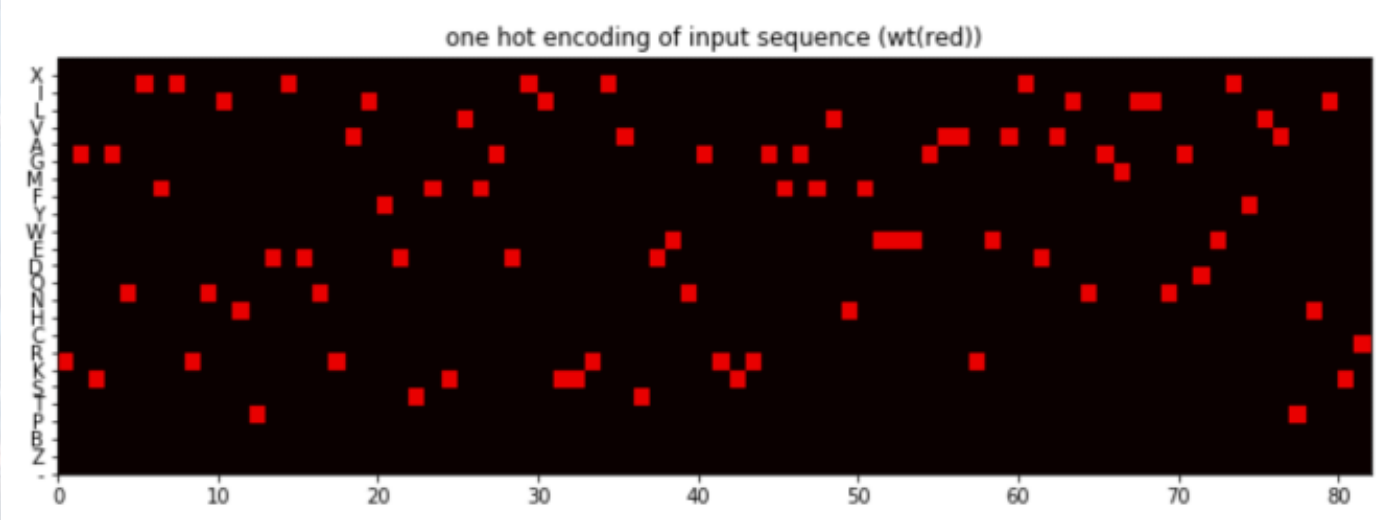
motif finding / sequence analysis



one – hot encoded NT or AA sequences can be interpreted as b/w images!



CNNs and Sequence Data

- barcodes are short DNA sequence for identifying species

- we want to see, if we can us a CNN for classification

- loading a so called ***fasta*** file `fasta.fas`

```
>BEISA025-19|Culex|COI-5P
AACATTATATTTCATTTTTGGTGCTTGAGCAGGAATAATTGGAACTTCTTTAAGTCTTCTTATTCG
AGCTGAATTAAGTCAACCAGGAGTTTTTATTGGGAATGATCAAATTTATAATGTAATTGTTACAGC
TCATGCTTTTATTATAATTTTTTTTATAGTAATACCTATTATAATTGGAGGATTTGGAAATTGATT
AGTTCCTTTAATACTAGGAGCTCCTGATATAGCTTTTCCTCGAATAAATAATATAAGATTTTGAAT
ACTTCCCCCCTCATTAACACTTCTACTTTCTAGTAGTATAGTAGAAAATGGAGCTGGTACAGGTTG
AACAGTATATCCTCCTCTTTCTTCTGGAACAGCTCATGCTGGAGCTTCTGTTGATTTAGCTATTTT
TTCTTTACATTTAGCCGGAATTTCTTCAATTTTAGGAGCTGTAAATTTTATTACTACTGTAATTAA
TATGCGATCTTCTGGTATTACCCTTGATCGAATACCTTTATTTGTTTGATCAGTTGTAATTACTGC
TATTCTTTTATTATTATCTCTTCCTGTTTTAGCTGGAGCTATTACTATATTATTAACAGATCGTAA
TTTAAATACTTCTTTTTTCGATCCTATTGGAGGAGGAGATCCTATTTTATATCAACATTTATTT
>BEISA121-19|Anopheles|COI-5P
AACATTATATTTTATTTTCGGTGCTTGAGCAGGAATAGTAGGAACTTCTTTAAGTATTCTTATTCG
```

true label

sequence as image

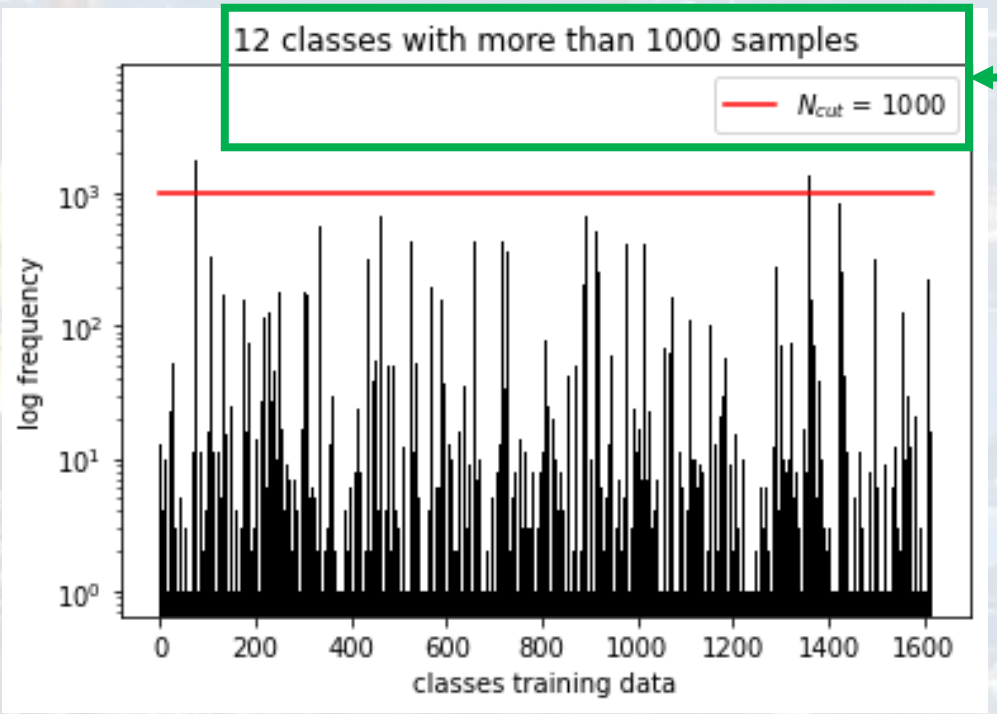
the data set:

- 86k samples
- 1.6k classes
- classes are not evenly distributed

→ picking only those with >1k samples (12 classes)

run the package `AnalyzBarcode3.py`

```
A = Analyzer()
```

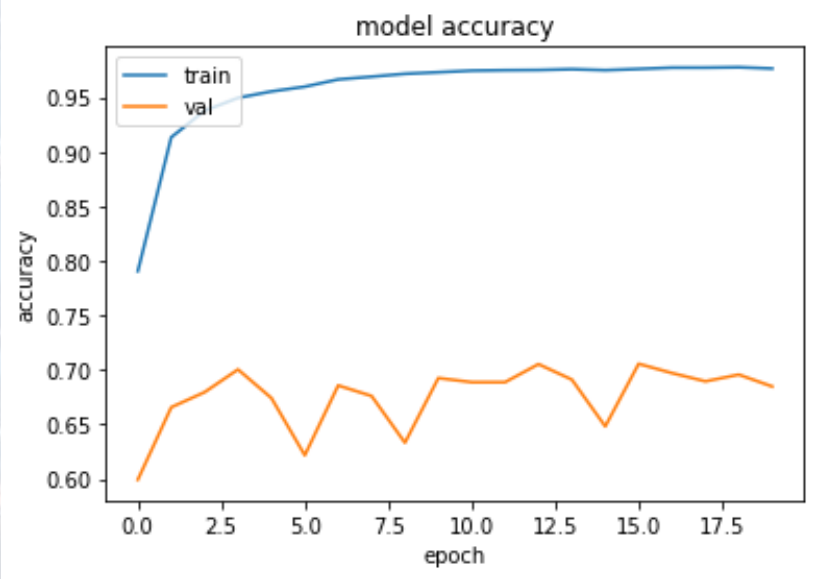


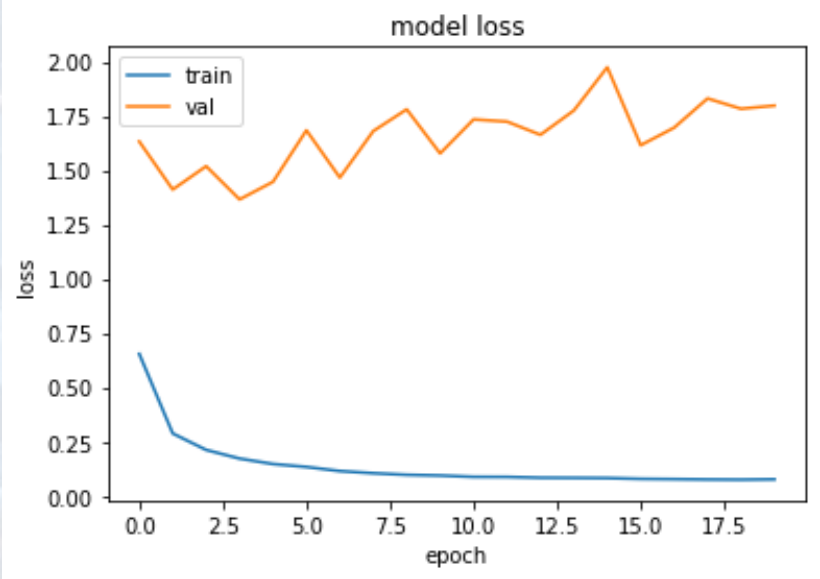
`__init__`
reads the data, one-hot encodes the nucleotides and passes only those samples where `Nsample > Ncut`

```
A.RunCNN()
```

runs very simple CNN

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 1256, 1, 24)	408
flatten (Flatten)	(None, 30144)	0
dense_2 (Dense)	(None, 84)	2,532,180
dense_3 (Dense)	(None, 12)	1,020



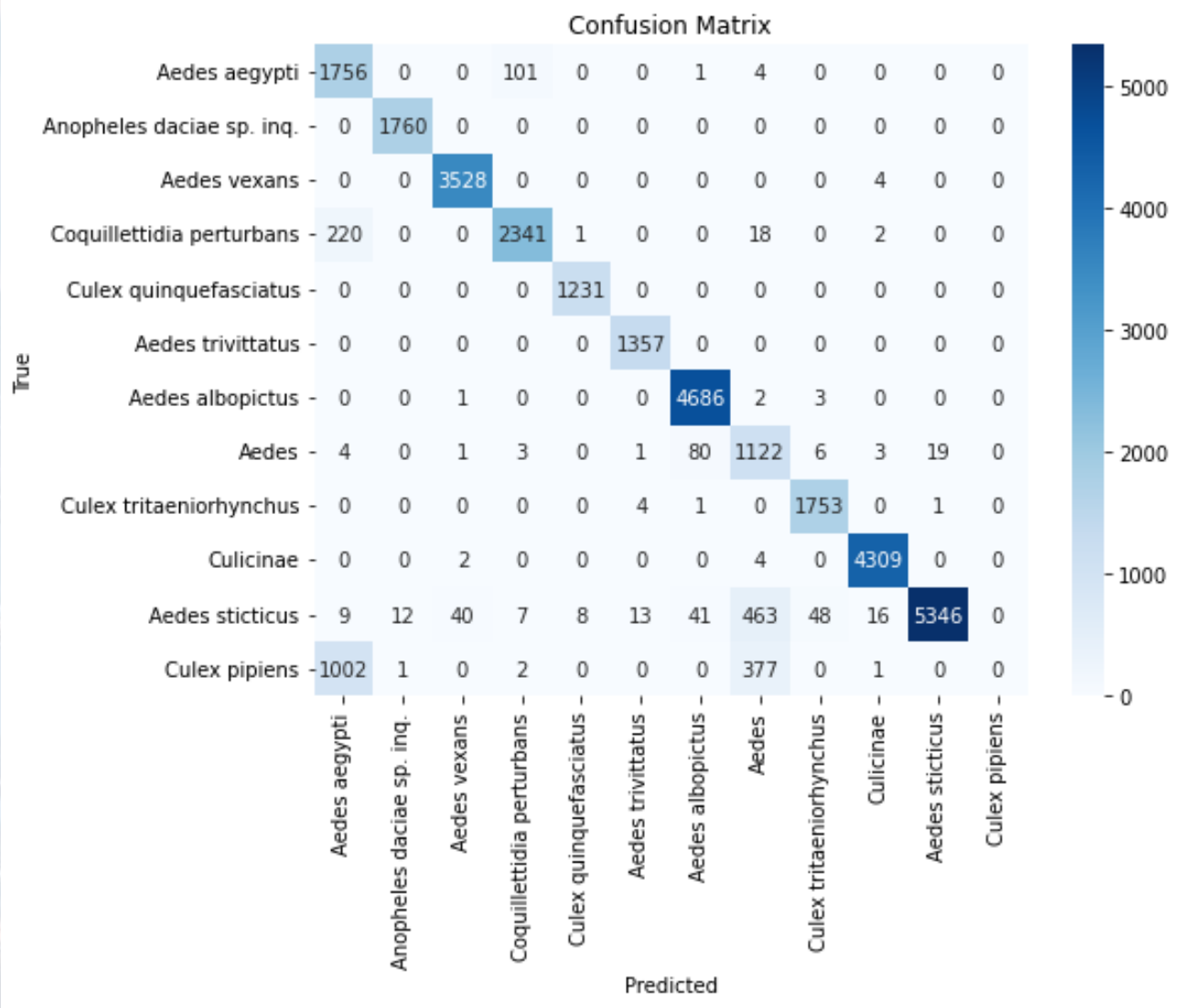


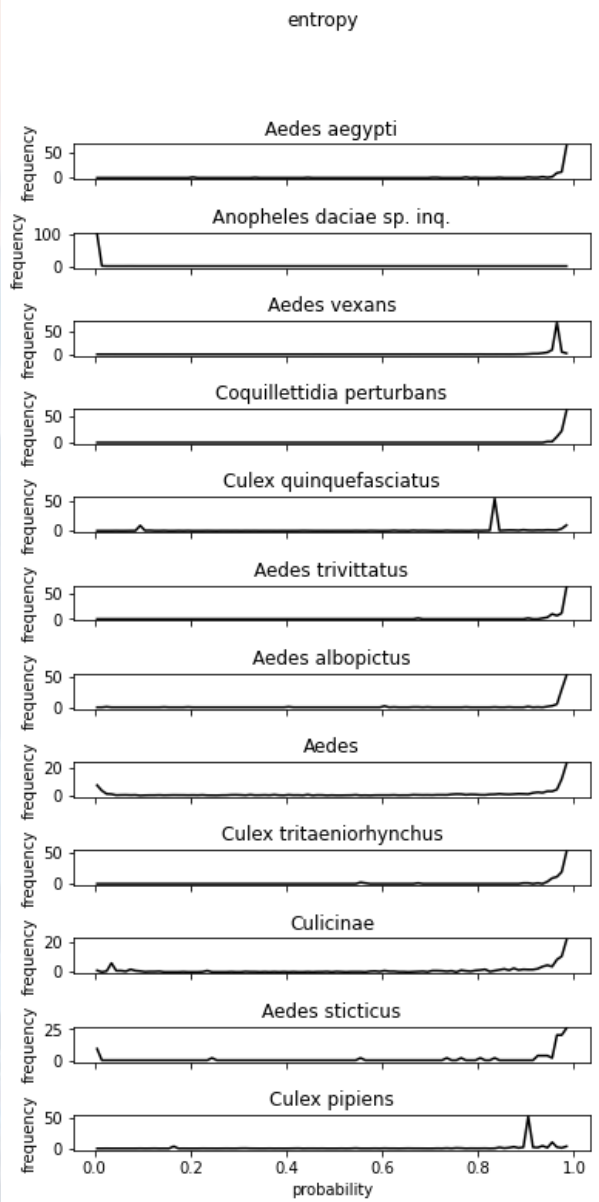
results are not great,
but it shows the principle

CNNs and Sequence Data

`A.EvalModel()`

evaluation **(note: here with training data)**



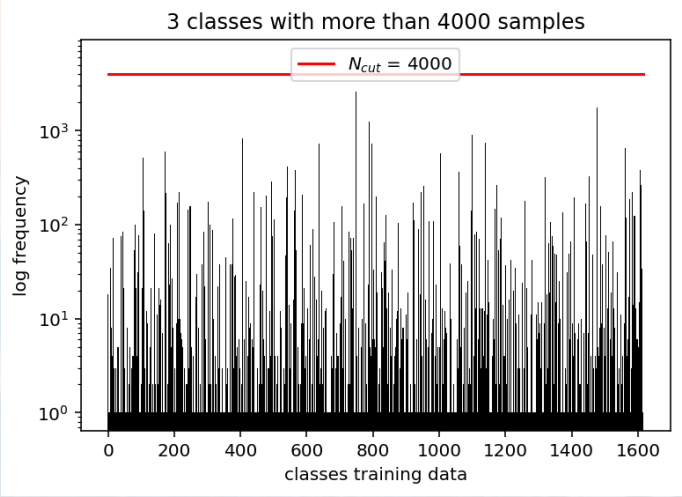


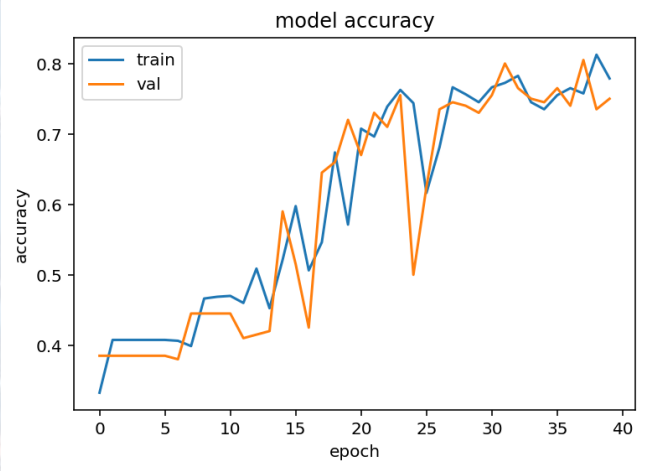
CNNs and Sequence Data

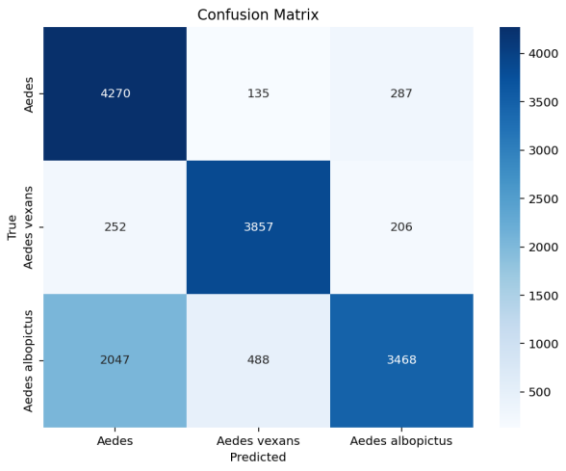
```
A.RunLSTM()
```

runs very simple LSTM (limit to 3 classes bco runtime)

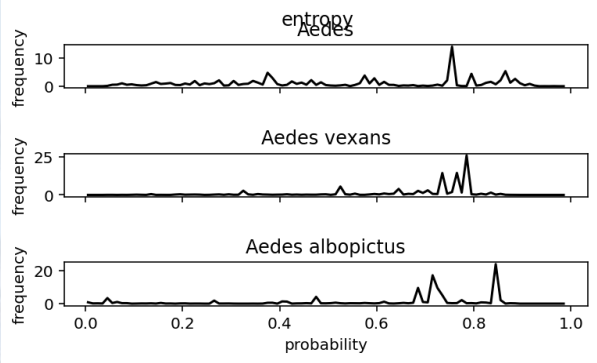
Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 100)	42,000
dense_4 (Dense)	(None, 3)	303







results are not great,
but better than before

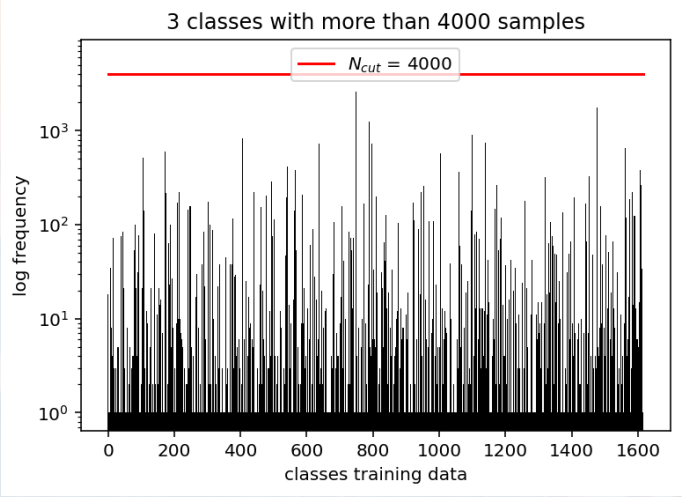


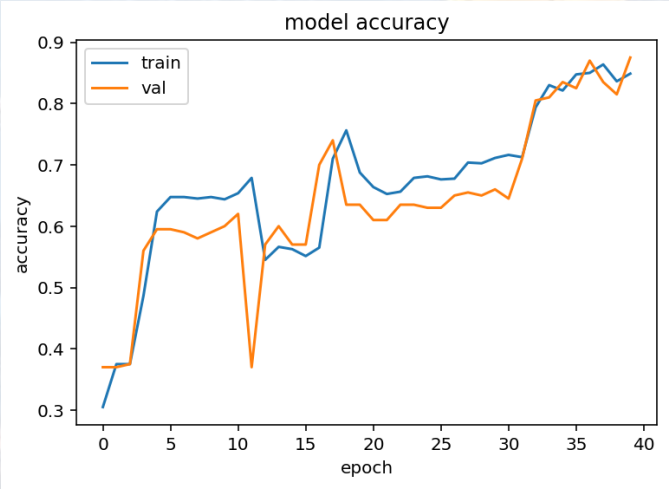
CNNs and Sequence Data

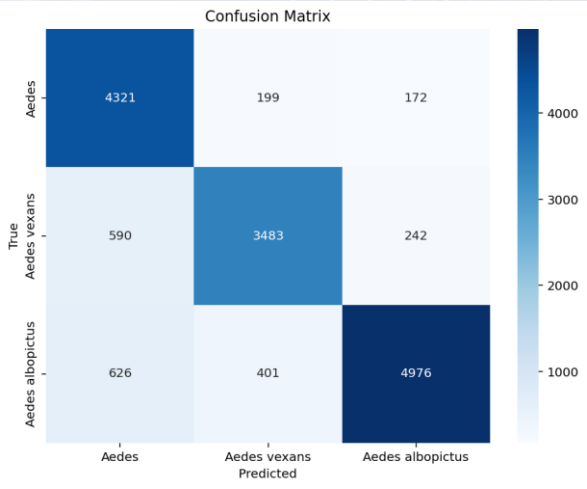
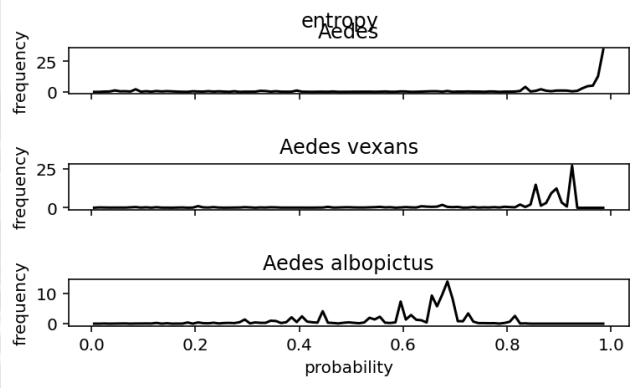
A.RunLSTMCNN()

runs very simple LSTM (limit to 3 classes bco runtime)

Layer (type)	Output Shape	Param #
conv1d (Conv1D)	(None, 498, 64)	832
max_pooling1d (MaxPooling1D)	(None, 249, 64)	0
lstm_1 (LSTM)	(None, 100)	66,000
dense_5 (Dense)	(None, 3)	303







results even better than before